

VIRAJ RAVINDRA BARAPATRE

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PROFESSIONAL SUMMARY

Highly analytical and research-oriented Computer Science graduate (Class of 2026) specializing in Quantitative Development, Data Analytics, and Artificial Intelligence. Verified elite-level logic performance with a **SIG CodeSignal Full Score (100th Percentile)** and advanced infrastructure expertise certified by **IIT Bombay**. Proven track record engineering automated data validation pipelines to process large-scale financial records, optimizing complex relational databases, and designing containerized deep learning applications.

TECHNICAL SKILLS

- **Languages:** Python (Pandas, NumPy, Scikit-Learn, TensorFlow, Keras), SQL (PostgreSQL, MySQL), C++, JavaScript/TypeScript, PHP, R, HTML5.
- **Data Science & AI:** Deep Learning, Computer Vision, Natural Language Processing, Predictive Modeling, Trend Analysis.
- **Engineering & Cloud:** Kubernetes (K8s), Docker, Cloud Computing (GCP, AWS, Azure), Terraform, Git/GitHub, REST APIs, Postman.
- **Analytics & Tools:** Quantitative Finance, Anomaly Detection, Data Pipelines (ETL/ELT), Advanced Excel, Business Intelligence.

PROFESSIONAL EXPERIENCE

TRUST FINTECH LIMITED

Data Science & Quantitative Intern

Nagpur, India
Dec 2025 – Present

- **Automated Data Pipelines:** Designed and deployed robust automated validation and transformation pipelines to process over 50,000 financial records daily, reducing pipeline latency and increasing total system operational efficiency by **35%**.
- **RDBMS Infrastructure Optimization:** Managed complex database architectures to guarantee **99.9% data integrity** for unallocated transaction tracing, optimizing query layouts and streamlining data retrieval performance.
- **Statistical Dashboarding:** Generated custom analytical reports and ad-hoc dashboards using advanced query scripts, converting fragmented datasets into decision-ready insights for executive leadership.

TECHNICAL PROJECTS

CAMPUSEYE: AI-AUGMENTED SURVEILLANCE PLATFORM *Next.js, Kubernetes, Gemini API, OpenCV*

- **Real-Time Computer Vision:** Architected an end-to-end, real-time campus surveillance system using computer vision pipelines and multi-modal Vision-Language Model (VLM) APIs for automated digital video analysis and threat detection.
- **Distributed Workloads:** Orchestrated high-frequency model microservices using **Kubernetes** to maintain real-time data telemetry, ensuring high-availability and zero-latency event alerting.
- **Prompt Engineering:** Formulated and iteratively optimized complex contextual prompt frameworks to refine unstructured image vectors, accelerating threat classification accuracy by **20%**.

DYNAMIC PRICING SIMULATOR

Python, TensorFlow, Keras, NumPy

- **Market Elasticity Modeling:** Built a deep neural network leveraging TensorFlow and Keras to simulate shifting market conditions and predict price fluctuations, validating performance to achieve a strict **RMSE of 13.36**.
- **Adaptive Feedback Loops:** Engineered recursive evaluation loops to track and analyze real-time buyer behavior sequences, systematically optimizing model response weights.

AML-SENTINEL-AI: ANOMALY DETECTION ENGINE

Python, Scikit-Learn, Pandas

- **Predictive Intelligence:** Developed an unsupervised machine learning engine utilizing the **Isolation Forest** algorithm to capture and flag non-linear structural risks in financial transaction records.
- **Rigorous Backtesting:** Performed exhaustive statistical inference and data mining workflows to optimize system thresholds and minimize false-positive alert rates.

ACHIEVEMENTS

- **SIG CodeSignal:** Achieved a perfect 600/600 score, validating world-class speed in core algorithmic reasoning, data structures, and advanced computational logic.
- **IIT Bombay RDBMS PostgreSQL and PHP and MySQL:** Professional certification specializing in advanced relational database design, query optimization, and secure backend data management.
- **Oracle OCI 2025 AI Foundations Associate:** Professional certification demonstrating mastery over enterprise cloud infrastructures, large-scale model orchestration, and machine learning governance.
- **Codecademy Professional Certifications:** Completed career-track certifications in Advanced Data Science, Computer Science, and SQL Data Engineering.

EDUCATION

B.TECH IN COMPUTER SCIENCE & ENGINEERING

G.H. Rasoni College of Engineering (GHRCE), Nagpur, India
Specialization: Artificial Intelligence

Graduating June 2026
CGPA: 8.01/10.00